Graph Convolutional Sparse Subspace Coclustering With Nonnegative Orthogonal Factorization for Large Hyperspectral Images

Nan Huang, Liang Xiao, *Member, IEEE*, Jianjun Liu, *Member, IEEE*, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— Sparse subspace clustering (SSC) is a representative data clustering paradigm that has been broadly applied in the unsupervised classification of hyperspectral images (HSIs). Existing SSC methods usually produce a subspace affinity matrix between representations of hyperspectral pixels first, followed by spectral clustering for the affinity matrix. To this end, the separated framework fails to exploit the dualities contained in both features and pixels or higher order entities at the same time, and thus, it is difficult to compute coclusters simultaneously. In addition, SSC methods often require expensive computational consumption and memory capacity to approximate the spectral decomposition of the affinity matrix, thus hindering the applicability of SSC methods for large HSIs. To overcome these limitations, we propose a novel graph convolutional sparse subspace coclustering (GCSSC) model with nonnegative orthogonal factorization for large HSIs in which affinity matrix learning and spectral coclustering are integrated into a unified optimizing model to obtain the optimal clustering results. Specifically, to form a more compact self-representation, the superpixel-based adaptive dictionary construction strategy is proposed instead of the global dictionary to precisely represent the pixels. To explore the spatial–contextual and spectral neighboring characteristics between dictionary atoms, graph convolution is incorporated into the dictionary atoms to aggregate the local neighborhood information, and the affinity matrix in the proposed coclustering framework is constructed under a joint sparsity constrained representation model. To reduce high computational consumption and memory capacity, a nonnegative orthogonal factorization constraint is proposed to offer an alternative spectral clustering for hyperspectral pixels and dictionary atoms simultaneously. The clustering performance of the proposed method is evaluated for three classical HSIs, and the experimental results illustrate that the proposed method is memory and computationally efficient and outperforms the state-of-the-art HSI clustering methods.**

***Index Terms*— Affinity matrix, nonnegative orthogonal factorization, sparse subspace clustering (SSC), spectral coclustering.**

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NOMENCLATURE

Notations	*Description*
N	Number of hyperspectral pixels.
D	Number of bands.
W	Number of dictionary atoms.
K	Number of clusters.
$\mathbf{Y}$	Matrix of hyperspectral image.
$\mathbf{y}_i$	ith column of $\mathbf{Y}$ (the ith hyperspectral pixel).
$\mathbf{f}_k$	Probability of all dictionary atoms associated with the kth cluster.
$\mathbf{g}_k$	Probability of all hyperspectral pixels associated with the kth cluster.
$\mathbf{S}_i$	ith superpixel.
$\lceil \mathbf{S}_i \rceil$	Number of the member of the ith superpixel.
$\mathbf{X}$	Sparse coefficient matrix.
$\lvert\mathbf{X}\rvert$	Nonnegative matrix of $\mathbf{X}$.
$\mathcal{G} = (\mathcal{V}, \varepsilon)$	Graph $\mathcal{G}$ with nodes set $\mathcal{V}$ and edges set ε.
$\mathbf{A}$	Adjacency matrix of $\mathcal{G}$.
$\mathbf{D}$	Diagonal degree matrix of $\mathbf{A}$.
$\mathbf{L}$	Graph Laplacian of $\mathcal{G}$.
$\mathbf{L}_{sym}$	Symmetric normalized graph Laplacian of $\mathcal{G}$.
$\tilde{\mathbf{Y}}$	Matrix of dictionary atoms.
$\mathbf{M}_1$	Matrix of superpixel-level spatial context.
$\mathbf{M}_2$	Matrix of spectral neighborhood.
$\mathbf{F}$	Indicator matrix of the dictionary atoms clusters.
$\mathbf{G}$	Indicator matrix of the hyperspectral pixels clusters.
$\mathbf{F}_{ik}$	Probability of the ith dictionary atom belonging to the kth cluster.
$\mathbf{G}_{jk}$	Probability of the jth hyperspectral pixel belonging to the kth cluster.
$\mathbf{\Lambda}_1$, $\mathbf{\Lambda}_2$	Lagrangian multiplier.

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$\mathcal{N}(\bullet)$	Neighborhood of $\bullet$.
$\lambda_{\max}$	Largest eigenvalue.
λ_k	kth eigenvalue.
$*$	Convolution operator.
$\odot$	Elementwise matrix multiplication.

I. Introduction

HYPERSPECTRAL image (HSI) clustering is a fundamental technique for HSI applications and has become a vital tool in the field of remote image analysis for forest inventors, environmental monitoring, precision agriculture, and land-use mapping [1]–[3]. In an unsupervised setting, clustering of HSIs assigns the unique land-cover classes to each hyperspectral pixel with no prior label information. Compared with supervised classification, clustering of HSIs is a more challenging task since more powerful unsupervised feature learning is required to find inherent relationships and group structures in the datasets. For HSIs, various clustering methods with different mechanisms have been proposed in recent years, and they can be roughly divided into nongraph-based clustering methods and graph-based clustering methods [4].

Nongraph-based clustering methods generally achieve good clustering accuracies when the intrinsic structures of the hyperspectral data are simple and pixels are not disturbed by noise. For instance, k-means [5], fuzzy c-means [6], and kernel-based fuzzy clustering [7] iteratively update the centroids and assign the clusters for hyperspectral pixels until the partition error between centroids and pixels meets the tolerance condition. Density-based spatial clustering of applications with noise (DBSCAN) [8] and ordering points to identify the clustering structure (OPTICS) [9] seek the locating regions of higher density to construct the clusters, and the density regions are calculated by the number of pixels within some radius. Agglomerative hierarchical clustering [10] and divisive hierarchical clustering [11] seek to build a clustering hierarchy by recursively segmenting the hyperspectral pixels in bottom-up or top-down methods until the tolerance condition is met.

Graph-based clustering methods generally convert data clustering into graph-cut problems by constructing the similarity among hyperspectral pixels [4], [12]. The goal of graph-based clustering is to divide a weighted graph into disjoint subsets and minimize the sum of the weights of the edges connecting the disjoint subsets. As one of the typical clustering methods, self-expressive subspace clustering uses self-representation models to construct a favorable affinity matrix by imposing a prior or regularization on the representations. In general, they assume that the hyperspectral pixels belonging to the same land-cover class lie in the same subspace. Under the subspace-based graph clustering framework, sparse and collaborative representation methods were introduced to propose new data clustering algorithms. For instance, Elhamifar and Vidal [13] and [14] proposed sparse subspace clustering (SSC) with l_0- or l_1-norm regularization. Zhai *et al.* [15] proposed integrating collaborative subspace clustering (CSC) and total variation into a unified framework under a local adaptive dictionary constraint. In [16], the rich spatial properties and high spectral correlation of HSIs were brought into the SSC framework to compute a more robust representation. Another typical method is scalable subspace clustering methods, which resolves the scalability issue for SSC. For instance, Peng *et al.* [17] proposed an out-of-sample extension of SSC to cluster large-scale data. Traganitis and Giannakis [18] proposed a sketch subspace clustering framework to reduce the computational complexity of subspace clustering by using random projection compressing a data matrix from both dimensions. In [19], kernel SSC was proposed by incorporating spatial and spatial max pooling information for the nonlinear structure of HSIs. In [20], a robust subspace clustering, which recasts the self-expressiveness property of the data into the non-Euclidean domain, was proposed by incorporating graph embedding into the self-representation coefficient matrix. In [21], a structure-enforced dictionary learning-based subspace clustering method was proposed to reduce the dimensions of dictionaries and coefficient matrices. In [22], low-rank and sparse-based robust subspace representations were presented for high-dimensional corrupted data. However, these graph-based clustering methods generally partition samples according to their features or partition features according to their distribution on samples and fail to explore the duality and co-occurrence properties between samples and features.

To overcome the limitations of the graph-based clustering methods mentioned above, coclustering methods have been proposed to exploit the co-occurrence structure between samples and features in recent decades [23], [24]. The goal of coclustering is to compute clusters of data samples and the related features concurrently. In general, coclustering constructs a graph Laplacian by using a spectral graph represented by the edge weight matrix, which represents the correlation between feature-sample pairs to compute low-dimensional embedding [25]. The majority of the spectral coclustering framework follows two stages: 1) constructing the adjacency matrix to describe the similarity between samples and features and 2) performing centroid-based methods on the left and right singular vectors of the Laplacian matrix of the adjacency matrix to get the final clustering results. However, in many practical applications, spectral coclustering follows two separate stages to partition the samples and features into clusters, and it is sensitive to the affinity matrix because the learned affinity graph matrix cannot guarantee to obtain the global optimal performance for clustering and results in suboptimal results. Moreover, spectral coclustering requires singular value decomposition (SVD) for the normalized affinity matrix, leading to problems such as high computational consumption and beyond the memory capacity of a CPU or GPU device, especially for large-scale HSIs [26]. Recently, nonnegative matrix factorization (NMF) has attracted great interest as a clustering method, and it was reported to obtain competitive performance compared with most advanced clustering algorithms [27]–[29]. Moreover, Chris *et al.* [28] proved some equivalence between NMF, k-means, and spectral-based clustering, while, in [30], NMF with additional orthogonal constraints was proposed to offer an efficient alternative solution for spectral clustering.

Inspired by the aforementioned study, we propose a graph convolutional sparse subspace coclustering (GCSSC) method with nonnegative orthogonal factorization for large HSIs.

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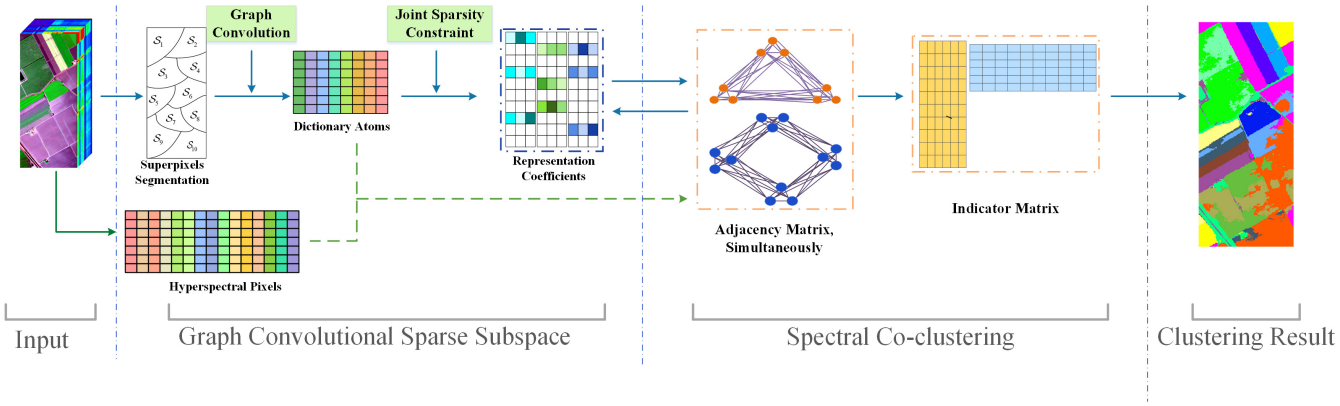


Fig. 1. Framework of the proposed coclustering method.

As illustrated in Fig. 1, the framework of the proposed coclustering method takes two aspects into account: the graph convolutional sparse subspace representation and spectral clustering procedure. However, different from the traditional spectral coclustering framework in separated implementation, the proposed coclustering method can be derived by a unified optimizing model, which integrates affinity matrix learning and spectral coclustering. The rationale behind the model is to impose NMF with additional orthogonal constraints on a graph convolutional sparse subspace model. Since the NMF-based method can provide an efficient iterative solution for spectral-based clustering, as mentioned above in [28], it motivates us to integrate these NMF-constraints into the final optimizing model. Because the optimization of the proposed model can be solved with an efficient iterative scheme without SVD of the affinity matrix, the derived coclustering algorithm is memory and computationally efficient for large HSIs.

The main contributions of this article are summarized as follows.

1) A graph convolutional sparse subspace model is proposed to build the coclustering mechanism in which the graph convolution constraint is incorporated into adaptive dictionary atoms. Differing from self-expressive subspace learning, the proposed approach constructs an adaptive dictionary instead of a global dictionary to precisely represent the hyperspectral pixels. Moreover, to explore the spatial–contextual and spectral neighboring information of dictionary atoms, graph convolution is further used to aggregate the local neighborhood information. To better represent the relationship pairs of pixel–atom, a joint sparsity representation is incorporated into the adaptive dictionary atoms under the $l_{2,1}$-norm regularization to capture the explicit correlation information between dictionary atoms and hyperspectral pixels.
2) We present a unified optimizing model for GCSSC in which the graph convolutional sparse subspace model and nonnegative orthogonal factorization constraints are integrated to boost the self-expressive power and coclustering performance.
3) With efficient multiplicative update rules (MUR), we propose a novel memory and computationally efficient iterative algorithm for the coclustering of large HSIs. Extensive experimental results on three typical HSIs illustrate that the proposed coclustering model performs more effectively than existing clustering methods for large HSIs.

The rest of this article is organized as follows. Section II describes SSC and graph convolution in detail. Section III presents the framework of coclustering for large HSIs. Section IV describes and analyzes the experimental results. Finally, Section V presents the conclusion of this article.

II. Preliminaries and Related Works

A. Notations

Throughout this article, sets, matrices, and vectors are denoted by calligraphy symbols, uppercase bold letters, and lowercase bold letters, respectively. $\mathbf{Y} = \left[\mathbf{y}_1, \ldots, \mathbf{y}_i, \ldots, \mathbf{y}_N\right] \in R^{D\times N}$ denotes 2-D HSI data, with each column denoting a hyperspectral pixel, where N denotes the number of hyperspectral pixels and D denotes the number of bands. Let $\mathcal{G} = (\mathcal{V}, \varepsilon, \mathbf{A})$, where $\mathcal{V}$ denotes the node set, ε denotes the edge set, and $\mathbf{A}$ denotes the weight (or affinity) matrix. $\mathbf{D}$ denotes the diagonal degree matrix of the affinity matrix, i.e., $\mathbf{D}_{ii} = \sum_j \mathbf{A}_{ij}$. $\mathbf{L} = \mathbf{D} - \mathbf{A}$ denotes the graph Laplacian, and its symmetric normalized version is $\mathbf{L}_{\text{sym}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$. In this article, for any matrix $\mathbf{X}$, $\mathbf{x}_i$ denotes the ith column vector of $\mathbf{X}$, and $\mathbf{X}_{ij}$ denotes the (i, j) element of $\mathbf{X}$. $\text{Tr}(\mathbf{X})$ and $\mathbf{X}^{\text{T}}$ denote the trace and transpose of matrix $\mathbf{X}$, respectively. $\mathbf{I}$ denotes the identity matrix. For convenience of discussion, we summarize the important notations used in this article in Nomenclature.

B. SSC for HSIs

In this section, we introduce the SSC for the HSIs. In general, the hyperspectral pixels can be expressed by a union of linear subspaces with each subspace corresponding to a certain land-cover type. The goal of SSC is to explore the self-representation property of HSIs and represent the pixel by a linear combination of the others [31]. The SSC model calculates the similarity among pixels by solving the following self-representation optimization problem:

$$\begin{aligned} &\min_{\mathbf{X}} \|\mathbf{Y} - \mathbf{Y}\mathbf{X}\|_F^2 + \alpha\|\mathbf{X}\|_1 \\ &\text{s.t.} \quad \text{diag}(\mathbf{X}) = 0 \end{aligned} \tag{1}$$

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where $\mathbf{X} \in R^{N\times N}$ denotes the self-expressive sparse coefficient matrix with the constraint of the diagonal elements to be zeroes, to avoid the trivial solutions of each pixel being represented as a linear combination of itself.

After computing the sparse coefficient of all pixels in (1), the affinity matrix can be constructed as follows:

$$\mathbf{A} = |\mathbf{X}| + |\mathbf{X}|^{\mathrm{T}} \tag{2}$$

where $\mathbf{A} \in R^{N\times N}$ denotes the adjacency matrix, $\mathbf{A}_{ij}$ denotes the similarity between the pixels i and j, and $|\mathbf{X}|$ denotes the nonnegative matrix of $\mathbf{X}$. Finally, the subspace clustering results can be achieved by performing spectral clustering for the adjacency matrix. The entire SSC algorithm is described in Algorithm 1.

Algorithm 1 SSC for HSIs

Input: $\mathbf{Y} \in R^{D\times N}$: hyperspectral image, K: number of clusters.

1: Perform the sparse encode to obtain the sparse coefficient matrix $\mathbf{X}$.

2: Construct the adjacency matrix $\mathbf{A}$.

3: Compute the symmetric normalized Laplacian matrix $\mathbf{L}_{sym} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$, where $\mathbf{D}_{ii} = \sum_j \mathbf{A}_{ij}$.

4: Compute the eigenvector matrix $\mathbf{U} \in R^{N\times K}$, which is the first K eigenvectors of $\mathbf{L}_{sym}$.

5: Employ k-means for the row of $\mathbf{U}$ to achieve the final clustering results.

Output: The clustering results of hyperspectral pixels $\mathbf{Y}$.

C. Spectral Graph Convolution in HSIs

From the perspective of the convolution theorem, a spectral graph convolution is defined as the multiplication of a hyperspectral pixel $\mathbf{y}_i$ with a filter $g_\theta = \mathrm{diag}(\theta)$ parameterized by θ in the Fourier domain [32], and it can be described as follows:

$$g_\theta * \mathbf{y}_i = \mathbf{U} g_\theta \mathbf{U}^{\mathrm{T}} \mathbf{y}_i \tag{3}$$

where $\mathbf{U}$ is the eigenvector matrix of symmetric normalized Laplacian matrix $\mathbf{L}_{\mathrm{sym}} = \mathbf{U}\Theta\mathbf{U}^{\mathrm{T}}$, Θ is the diagonal matrix of its eigenvalues, and $\mathbf{U}^{\mathrm{T}}\mathbf{y}_i$ is the graph Fourier transform of $\mathbf{y}_i$. Through the above analysis, we can view g_θ as a function of the eigenvalues of $\mathbf{L}_{\mathrm{sym}}$, denoted as $g_\theta(\Theta)$. To decrease the computational complexity in the eigendecomposition of $\mathbf{L}_{\mathrm{sym}}$, Hammond *et al.* [33] applied the kth order truncated expansion of the Chebyshev polynomials to approximate $g_\theta(\Theta)$. Therefore, (3) can be reformulated as

$$g_\theta * \mathbf{y}_i \approx \sum_{l=0}^{L} \theta'_l \mathbf{T}_l\big(\tilde{\mathbf{L}}\big)\mathbf{y}_i \tag{4}$$

where $\tilde{\mathbf{L}} = (2/\lambda_{\max})\mathbf{L}_{\mathrm{sym}} - \mathbf{I}$, $\lambda_{\max}$ is the largest eigenvalue of $\mathbf{L}_{\mathrm{sym}}$, $\mathbf{T}_l\big(\tilde{\mathbf{L}}\big)$ is the Chebyshev polynomial corresponding to $\tilde{\mathbf{L}}$, and θ'_l is the Chebyshev coefficient.

By constraining $L = 1$ and approximating $\lambda_{\max} \approx 2$ [34], (4) can be simplified as follows:

$$g_\theta * \mathbf{y}_i \approx \theta\big(\mathbf{I} + \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\big)\mathbf{y}_i. \tag{5}$$

III. Proposed Method

In this section, we introduce the proposed coclustering method for large HSIs in three parts: graph convolutional sparse subspace, nonnegative orthogonal factorization constraints, and a unified framework of subspace learning and spectral clustering. More details are described as follows.

A. Superpixel-Based Adaptive Dictionary Atom Construction

In (1), the hyperspectral pixels are expressed by a linear combination of all pixels [35]. However, the pixels are highly correlated with each other, and those belonging to the same class share similar spectral characteristics. In other words, similar atoms in the global dictionary have a negative effect on the representation of hyperspectral pixels. To better represent the intrinsic features and reduce the computational time, the adaptive selection of dictionary atoms becomes distinctly important to precisely represent the pixels, and it can be described as follows:

$$\begin{aligned} &\min_{\tilde{\mathbf{Y}},\mathbf{X}} \big\|\mathbf{Y} - \tilde{\mathbf{Y}}\mathbf{X}\big\|_F^2 + \alpha\|\mathbf{X}\|_1 \\ &\text{s.t. } \mathrm{diag}(\mathbf{X}) = 0 \end{aligned} \tag{6}$$

where $\tilde{\mathbf{Y}} = [\tilde{\mathbf{y}}_1, \ldots, \tilde{\mathbf{y}}_W] \subseteq \mathbf{Y}$ is the subset of hyperspectral pixels, and $W \in \{1, \ldots, N\}$ denotes the number of atoms in the subset. The key to this method involves utilizing a small number of representative atoms to capture the intrinsic characteristics of HSIs. In this article, we propose anchor-based dictionary representation to seek W anchors from global dictionary atoms.

Specifically, we first perform principal component analysis and retain the first three principal components to capture the most important HSI information and then perform superpixel segmentation via entropy rate superpixel segmentation (ERS) [36], [37]. Next, we obtain a number of spatial neighborhoods grouped by the adaptive superpixel patches (see Fig. 2 for details). Let $\{\mathbf{S}_1 \ldots, \mathbf{S}_i, \ldots \mathbf{S}_W\}$ denote the set of superpixels and $\mathbf{Y}_{\mathbf{S}_i} = [\mathbf{y}_{i,1}, \ldots, \mathbf{y}_{i,\lceil \mathbf{S}_i \rceil}]$ denote the member of superpixel $\mathbf{S}_i$, where W is the number of superpixels and $\mathbf{y}_{i,\lceil \mathbf{S}_i \rceil}$ denotes the $\lceil \mathbf{S}_i \rceil$th hyperspectral pixel in the superpixel $\mathbf{S}_i$. We utilize the center of superpixel $\mathbf{S}_i$ to construct the anchors of hyperspectral pixels, and the anchor construction model can be formulated as follows:

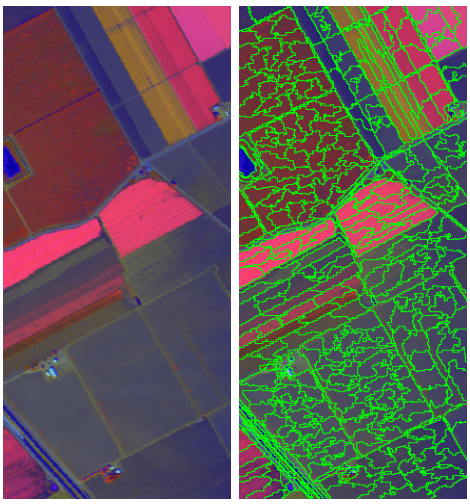
$$\tilde{\mathbf{y}}_i = \frac{1}{\lceil \mathbf{S}_i \rceil} \sum_{j=1}^{\lceil \mathbf{S}_i \rceil} \mathbf{y}_{i,j} \tag{7}$$

where $\tilde{\mathbf{y}}_i$ denotes the ith anchor. Compared with the anchors of rand selection or k-means method, the proposed anchor construction method utilizes the homogeneous regions of HSIs, and it is more representative by using the superpixel centers.

B. Graph Convolutional Sparse Subspace

We clearly observe that (6) fails to exploit the intrinsic structural and geometric relationship between dictionary atoms. For HSIs, the dictionary atoms in spatial and spectral neighborhoods have a high probability of sharing the same spectral

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(a) (b)

Fig. 2. Segmentation result using ERS for Salinas. (a) False-color image. (b) Superpixel map.

information, and the dictionary atoms in the same land-cover class should also be similar. In other words, dictionary atoms in the spatial context and same land-cover class should be very similar according to the collected working mechanism of the spatial–spectral characteristics. Thus, a graph convolution constraint is useful to incorporate the spatial–contextual and spectral neighboring information and improve the expressing ability of dictionary atoms, and the graph convolution can be called the local neighborhood aggregation function. The goal of combining graph convolution and dictionary atoms is to utilize graph convolution learning a robust dictionary by considering spatial contextual and node neighborhoods. The dictionary atoms and representation learning model can be described as follows:

$$\begin{aligned} &\min_{\mathbf{X}} \left\|\mathbf{Y}-\tilde{\mathbf{Y}}\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\mathbf{X}\right\|_F^2+\alpha\|\mathbf{X}\|_1 \\ &\text{s.t.} \quad \operatorname{diag}(\mathbf{X})=0 \end{aligned} \tag{8}$$

where $\mathbf{A}$ denotes the adjacency matrix of adaptive dictionary atoms $\tilde{\mathbf{Y}}$. In (8), $\mathbf{A}$ is calculated by considering the superpixel-level spatial contextual and node neighborhood features, and it can be described as follows:

$$\mathbf{A}=(\mathbf{M}_1+\mathbf{M}_2)\odot\mathbf{W} \tag{9}$$

where $\mathbf{M}_1$ denotes the superpixel-level spatial context, $\mathbf{M}_2$ denotes the spectral neighborhood, $\mathbf{W}_{ij}=\exp(-\|\tilde{\mathbf{y}}_i-\tilde{\mathbf{y}}_j\|^2/2\sigma^2)$ denotes the pairwise relationship between dictionary atoms, σ is a scale parameter that is regarded as the average value of all pairwise distances [38], and $\odot$ denotes the elementwise product. Here, we define the matrices of $\mathbf{M}_1$ and $\mathbf{M}_2$ in detail as follows:

$$(\mathbf{M}_1)_{ij}=\begin{cases}1, & \text{if } \mathbf{S}_i\in\mathcal{N}(\mathbf{S}_j)\\ 0, & \text{otherwise}\end{cases} \tag{10}$$

$$(\mathbf{M}_2)_{ij}=\begin{cases}1, & \text{if } \tilde{\mathbf{y}}_i\in\mathcal{N}(\tilde{\mathbf{y}}_j)\\ 0, & \text{otherwise}\end{cases} \tag{11}$$

where $\mathcal{N}(\mathbf{S}_j)$ denotes the neighborhood of $\mathbf{S}_j$ when they share a common boundary, and $\mathcal{N}(\tilde{\mathbf{y}}_j)$ denotes the spectral neighborhood of $\tilde{\mathbf{y}}_j$. Notably, $\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ can be regarded as a special linear graph convolution operation for dictionary atoms $\tilde{\mathbf{Y}}$.

For the HSIs, hyperspectral pixels within a superpixel $\mathbf{S}_i$ are assumed to be local homogeneous regions, and they can be expressed as linear combinations of common pattern atoms from the dictionary. Therefore, joint sparsity regularization is proposed to explore the local similarity of pixels within each superpixel, and the representation model can be expressed as follows:

$$\min_{\mathbf{X}}\left\|\mathbf{Y}-\tilde{\mathbf{Y}}\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\mathbf{X}\right\|_F^2+\alpha\sum_{i=1}^{W}\left\|\mathbf{X}_{\mathbf{S}_i}\right\|_{2,1} \tag{12}$$

where $\mathbf{X}=[\mathbf{x}_1,\ldots,\mathbf{x}_i,\ldots,\mathbf{x}_N]\in R^{W\times N}$ denotes the sparse coefficient from dictionary $\tilde{\mathbf{Y}}\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\in R^{D\times W}$, $\left\|\mathbf{X}_{\mathbf{S}_i}\right\|_{2,1}$ denotes the sum of the l_2-norm of the rows of $\mathbf{X}_{\mathbf{S}_i}$, and α denotes the tradeoff parameter between reconstruction error and priori.

C. Nonnegative Orthogonal Factorization Constraints

From (12), we observe that the locations of nonzero coefficients represent the dictionary atoms of each pixel, and their absolute values represent the corresponding weights of the dictionary atoms. Therefore, hyperspectral pixel clustering induces dictionary atom clustering, and vice versa.

In this article, we simultaneously cluster dictionary atoms and hyperspectral pixels by transforming the sparsity coefficients into the adjacency matrix. Here, the nonnegative joint sparsity coefficient $\mathbf{X}$ includes the associated relationship between row and column objects. We use spectral clustering to partition the hyperspectral pixels and dictionary atoms simultaneously. Specifically, letting the rows of $\mathbf{X}$ encode the associations related to the dictionary atoms $\tilde{\mathbf{Y}}\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$, we define $\mathbf{F}=[\mathbf{f}_1,\ldots,\mathbf{f}_k,\ldots,\mathbf{f}_K]\in R^{W\times K}$ as the indicator matrix of the dictionary atom clusters, where $\mathbf{f}_k$ denotes the probability that all dictionary atoms are associated with the kth cluster, $\mathbf{F}_{ik}$ denotes the probability that the ith dictionary atom belongs to the kth cluster, and $k\in\{1,\ldots,K\}$. Similarly, we also define $\mathbf{G}=[\mathbf{g}_1,\ldots,\mathbf{g}_k,\ldots,\mathbf{g}_K]\in R^{N\times K}$ as the indicator matrix of the hyperspectral pixel clusters since the columns of $\mathbf{X}$ contain the associations related to the hyperspectral pixels, where $\mathbf{g}_k$ denotes the probability that all hyperspectral pixels are associated with the kth cluster, and $\mathbf{G}_{jk}$ denotes the probability that the jth hyperspectral pixel belongs to the kth cluster. The objective function for simultaneous hyperspectral pixel and dictionary atom spectral clustering of $\mathbf{X}$ can be described as follows:

$$\begin{cases}\max\limits_{\mathbf{F}^{\mathrm{T}}\mathbf{F}=\mathbf{I},\mathbf{F}\geq 0}, & \operatorname{Tr}\left(\mathbf{F}^{\mathrm{T}}\mathbf{X}\mathbf{X}^{\mathrm{T}}\mathbf{F}\right)\\ \max\limits_{\mathbf{G}^{\mathrm{T}}\mathbf{G}=\mathbf{I},\mathbf{G}\geq 0}, & \operatorname{Tr}\left(\mathbf{G}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}\mathbf{G}\right).\end{cases} \tag{13}$$

Combing the two equations of (13), we obtain

$$\begin{aligned} &\max_{\mathbf{F},\mathbf{G}}\operatorname{Tr}\begin{pmatrix}\mathbf{F}\\ \mathbf{G}\end{pmatrix}^{\mathrm{T}}\begin{pmatrix}\mathbf{0} & \mathbf{X}\\ \mathbf{X}^{\mathrm{T}} & \mathbf{0}\end{pmatrix}\begin{pmatrix}\mathbf{F}\\ \mathbf{G}\end{pmatrix}\\ &\quad=2\operatorname{Tr}\left(\mathbf{F}^{\mathrm{T}}\mathbf{X}\mathbf{G}\right)\\ &\text{s.t.}\quad \mathbf{F}^{\mathrm{T}}\mathbf{F}=\mathbf{I},\quad \mathbf{G}^{\mathrm{T}}\mathbf{G}=\mathbf{I},\ \mathbf{F},\mathbf{G}\geq 0. \end{aligned} \tag{14}$$

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Next, we illustrate that (13) is equivalent to (14). First, the solution of (14) is achieved by the first K eigenvectors of $\begin{pmatrix} 0 & \mathbf{X} \\ \mathbf{X}^{\mathrm{T}} & 0 \end{pmatrix}\begin{pmatrix} \mathbf{f}_K \\ \mathbf{g}_K \end{pmatrix} = \lambda_K \begin{pmatrix} \mathbf{f}_K \\ \mathbf{g}_K \end{pmatrix}$, which can be further rewritten as $\mathbf{X}\mathbf{g}_K = \lambda_K \mathbf{f}_K$ and $\mathbf{X}^{\mathrm{T}}\mathbf{f}_K = \lambda_K \mathbf{g}_K$. Multiplying both sides of $\mathbf{X}\mathbf{g}_K = \lambda_K \mathbf{f}_K$ and $\mathbf{X}^{\mathrm{T}}\mathbf{f}_K = \lambda_K \mathbf{g}_K$ by $\mathbf{X}^{\mathrm{T}}$ and $\mathbf{X}$, respectively, we can obtain $\mathbf{X}^{\mathrm{T}}\mathbf{X}\mathbf{g}_K = \lambda_K \mathbf{X}^{\mathrm{T}}\mathbf{f}_K = \lambda_K \mathbf{g}_K$ and $\mathbf{X}\mathbf{X}^{\mathrm{T}}\mathbf{f}_K = \lambda_K \mathbf{X}\mathbf{g}_K = \lambda_K \mathbf{f}_K$, which are also the solutions to the equations in (13).

Furthermore, by adding two terms $\mathrm{Tr}(\mathbf{X}^{\mathrm{T}}\mathbf{X})$ and $\mathrm{Tr}(\mathbf{F}^{\mathrm{T}}\mathbf{F}\mathbf{G}^{\mathrm{T}}\mathbf{G})$, (14) can be rewritten as a matrix decomposition problem

$$\begin{aligned} \min_{\mathbf{F},\mathbf{G}} & -2\mathrm{Tr}(\mathbf{F}^{\mathrm{T}}\mathbf{X}\mathbf{G}) \\ & = \mathrm{Tr}(\mathbf{X}^{\mathrm{T}}\mathbf{X}) - 2\mathrm{Tr}(\mathbf{F}^{\mathrm{T}}\mathbf{X}\mathbf{G}) + \mathrm{Tr}(\mathbf{F}^{\mathrm{T}}\mathbf{F}\mathbf{G}^{\mathrm{T}}\mathbf{G}) \\ & = \left\|\mathbf{X} - \mathbf{F}\mathbf{G}^{\mathrm{T}}\right\|_{\mathbf{F}}^2 \\ \text{s.t.} \quad & \mathbf{F}^{\mathrm{T}}\mathbf{F} = \mathbf{I},\ \mathbf{G}^{\mathrm{T}}\mathbf{G} = \mathbf{I},\ \mathbf{F},\mathbf{G} \geq 0. \end{aligned} \tag{15}$$

It can be observed that the low-rank matrix $\mathbf{G}$ will cluster the hyperspectral pixels, and the low-rank matrix $\mathbf{F}$ will cluster the dictionary atoms. In addition, the clustering results can be directly obtained without the centroid-based method.

D. Unified Optimizing Model for GCSSC

From (12) and (13), we clearly observe that the coclustering method follows two stages to achieve the optimal clustering result for each stage separately but cannot ensure the global optimal solutions. In addition, the affinity matrix directly obtained from the original high-dimensional data space will seriously disturb the performance of HSI clustering because high-dimensional HSIs usually contain noise and redundancy. From this point of view, affinity matrix learning is integrated into the spectral coclustering model. We further proposed a unified optimization method to update these three matrices iteratively while fixing the other two. This method can be described as follows:

$$\begin{aligned} \min_{\mathbf{X},\mathbf{F},\mathbf{G}} & \left\|\mathbf{X} - \mathbf{F}\mathbf{G}^{\mathrm{T}}\right\|_F^2 + \gamma \left\|\mathbf{Y} - \tilde{\mathbf{Y}}\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\mathbf{X}\right\|_F^2 \\ & + \alpha \sum_{i=1}^{W} \left\|\mathbf{X}_{\mathbf{S}_i}\right\|_{2,1} \\ \text{s.t.} \quad & \mathbf{F}^{\mathrm{T}}\mathbf{F} = \mathbf{I}, \quad \mathbf{G}^{\mathrm{T}}\mathbf{G} = \mathbf{I},\ \mathbf{F},\mathbf{G} \geq 0. \end{aligned} \tag{16}$$

E. Optimizing Model for GCSSC

In this section, we use the MUR [39] to optimize (16). The objective function of (16) is nonconvex in terms of both $\mathbf{F}$ and $\mathbf{G}$, but it is convex in $\mathbf{F}$ or $\mathbf{G}$ individually. Specifically, we optimize the objective function for one variable while fixing other variables. We introduce a Lagrangian multiplier $\mathbf{\Lambda}_1$ to enforce $\mathbf{F}^{\mathrm{T}}\mathbf{F} = \mathbf{I}$ and another Lagrangian multiplier $\mathbf{\Lambda}_2$ to enforce $\mathbf{G}^{\mathrm{T}}\mathbf{G} = \mathbf{I}$. The Lagrangian function of (16) is described as follows:

$$\begin{aligned} J(\mathbf{X},\mathbf{F},\mathbf{G}) = & \left\|\mathbf{X} - \mathbf{F}\mathbf{G}^{\mathrm{T}}\right\|_F^2 \\ & + \gamma \left\|\mathbf{Y} - \tilde{\mathbf{Y}}\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\mathbf{X}\right\|_F^2 + \alpha \sum_{i=1}^{W} \left\|\mathbf{X}_{\mathbf{S}_i}\right\|_{2,1} \\ & + \mathrm{Tr}\left(\mathbf{\Lambda}_1\left(\mathbf{F}^{\mathrm{T}}\mathbf{F} - \mathbf{I}\right)\right) + \mathrm{Tr}\left(\mathbf{\Lambda}_2\left(\mathbf{G}^{\mathrm{T}}\mathbf{G} - \mathbf{I}\right)\right). \end{aligned} \tag{17}$$

In the following, the optimization of (17) can be divided into three subproblems.

1) Updating $\mathbf{X}$ and Fixed $\mathbf{F}$ and $\mathbf{G}$: The minimizing of (17) is equivalent to optimize the following problem:

$$\begin{aligned} J(\mathbf{X}) = & \left\|\mathbf{X} - \mathbf{F}\mathbf{G}^{\mathrm{T}}\right\|_F^2 + \gamma \left\|\mathbf{Y} - \tilde{\mathbf{Y}}\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\mathbf{X}\right\|_F^2 \\ & + \alpha \sum_{i=1}^{W} \left\|\mathbf{X}_{\mathbf{S}_i}\right\|_{2,1}. \end{aligned} \tag{18}$$

We can further rewrite the above equation as

$$\begin{aligned} J(\mathbf{X}) = & \left\| \begin{pmatrix} \sqrt{\gamma}\mathbf{Y} \\ \mathbf{F}\mathbf{G}^{\mathrm{T}} \end{pmatrix} - \begin{pmatrix} \sqrt{\gamma}\tilde{\mathbf{Y}}\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2} \\ \mathbf{I} \end{pmatrix}\mathbf{X} \right\|_F^2 \\ & + \alpha \sum_{i=1}^{W} \left\|\mathbf{X}_{\mathbf{S}_i}\right\|_{2,1}. \end{aligned} \tag{19}$$

This is the well-known joint sparse recovery problem, also called the multiple measurement vector (MMV) problem in compressed sensing. In this article, we adopt the regularized M-FOCUSS (regularized FOCal underdetermined system solver) strategy to solve (19) by iteratively updating each row of $\mathbf{X}_{\mathbf{S}_i}$ [40].

2) Updating $\mathbf{F}$ and $\mathbf{G}$, and Fixed $\mathbf{X}$: To update $\mathbf{F}$ and $\mathbf{G}$, the optimization of (17) can be expressed as follows:

$$\begin{aligned} J(\mathbf{F},\mathbf{G}) = & \left\|\mathbf{X} - \mathbf{F}\mathbf{G}^{\mathrm{T}}\right\|_F^2 + \mathrm{Tr}\left(\mathbf{\Lambda}_1\left(\mathbf{F}^{\mathrm{T}}\mathbf{F} - \mathbf{I}\right)\right) \\ & + \mathrm{Tr}\left(\mathbf{\Lambda}_2\left(\mathbf{G}^{\mathrm{T}}\mathbf{G} - \mathbf{I}\right)\right). \end{aligned} \tag{20}$$

The derivatives of J with respect to $\mathbf{F}$ and $\mathbf{G}$ are

$$\frac{\partial J}{\partial \mathbf{F}} = -2\mathbf{X}\mathbf{G} + 2\mathbf{F}\mathbf{G}^{\mathrm{T}}\mathbf{G} + 2\mathbf{F}\mathbf{\Lambda}_1 \tag{21}$$

$$\frac{\partial J}{\partial \mathbf{G}} = -2\mathbf{X}^{\mathrm{T}}\mathbf{F} + 2\mathbf{G}\mathbf{F}^{\mathrm{T}}\mathbf{F} + 2\mathbf{G}\mathbf{\Lambda}_2. \tag{22}$$

Using the Karush–Kuhn–Tuckre (KKT) conditions [41], we obtain the following updating for $\mathbf{F}_{ik}$ and $\mathbf{G}_{ik}$:

$$\mathbf{F}_{ik} \leftarrow \mathbf{F}_{ik}\left(\frac{(\mathbf{X}\mathbf{G})_{ik}}{\left(\mathbf{F}\mathbf{G}^{\mathrm{T}}\mathbf{G} + \mathbf{F}\mathbf{\Lambda}_1\right)_{ik}}\right) \tag{23}$$

$$\mathbf{G}_{jk} \leftarrow \mathbf{G}_{jk}\left(\frac{\left(\mathbf{X}^{\mathrm{T}}\mathbf{F}\right)_{jk}}{\left(\mathbf{G}\mathbf{F}^{\mathrm{T}}\mathbf{F} + \mathbf{G}\mathbf{\Lambda}_2\right)_{jk}}\right). \tag{24}$$

From the KKT condition, summing over k in (21), we can obtain $(\mathbf{F}^{\mathrm{T}}\mathbf{F}\mathbf{G}^{\mathrm{T}}\mathbf{G} - \mathbf{F}^{\mathrm{T}}\mathbf{X}\mathbf{G})_{ii} = (\mathbf{F}^{\mathrm{T}}\mathbf{F}\mathbf{\Lambda}_1)_{ii} = (\mathbf{\Lambda}_1)_{ii}$. For the off-diagonal of $\mathbf{\Lambda}_1$, we ignore the nonnegativity constraint and obtain $(\mathbf{\Lambda}_1)_{ii'} = (\mathbf{F}^{\mathrm{T}}\mathbf{F}\mathbf{G}^{\mathrm{T}}\mathbf{G} - \mathbf{F}^{\mathrm{T}}\mathbf{X}\mathbf{G})_{ii'}$. Combining the two together, we obtain $\mathbf{\Lambda}_1 = \mathbf{F}^{\mathrm{T}}\mathbf{F}\mathbf{G}^{\mathrm{T}}\mathbf{G} - \mathbf{F}^{\mathrm{T}}\mathbf{X}\mathbf{G}$. For $\mathbf{\Lambda}_2$, similarly, we obtain $\mathbf{\Lambda}_2 = \mathbf{G}^{\mathrm{T}}\mathbf{G}\mathbf{F}^{\mathrm{T}}\mathbf{F} - \mathbf{G}^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{F}$. It is worth noting that, once we obtain the indicator matrices $\mathbf{F}$ and $\mathbf{G}$, the cluster labels of the ith dictionary atom and jth hyperspectral pixel can be calculated as $\arg\max_k \mathbf{F}_{ik}$ and $\arg\max_k \mathbf{G}_{jk}$, respectively. The entire algorithm is described in detail in Algorithm 2.

F. Computational Complexity

In this section, we illustrate the computational complexity of the proposed GCSSC algorithm. As shown in Algorithm 2, superpixel segmentation via ERS is first proposed to obtain

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Algorithm 2 GCSSC With Nonnegative Orthogonal Factorization

Input: $\mathbf{Y} \in R^{D\times N}$: hyperspectral image, K: number of clusters.

1. Construct the adaptive dictionary atoms $\tilde{\mathbf{Y}}$ by Eq. 7.
2. Construct the graph convolution between dictionary atoms by the $\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$.

Repeat

3. Perform the sparse encoding under joint sparsity constraint to obtain the sparse coefficient matrix $\mathbf{X}$ in the Eq. 19.
4. Update the indicator matrix $\mathbf{F}$ of dictionary atoms by Eq. 23.
5. Update the indicator matrix $\mathbf{G}$ of hyperspectral pixels by Eq. 24.

Until the stopping criterion is satisfied

6. Compute the cluster label of jth hyperspectral pixels by $\arg\max_k \mathbf{G}_{jk}$, where $k \in \{1, \ldots, K\}$.

Output: The clustering results of hyperspectral pixels $\mathbf{Y}$.

the superpixels of HSIs. The lazy greedy [37] is introduced to solve this segmentation, and the computational complexity is $O(N^2 \log(N))$, where N denotes the number of hyperspectral pixels. Then, the adaptive dictionary atoms with graph convolution are constructed by (7) and (9). This needs a complexity of $O\left(ND + DW^2\right)$, where D denotes the number of bands, and W denotes the number of dictionary atoms. Finally, representation learning is incorporated into the spectral clustering framework. The proposed optimization problem is solved by updating the joint sparsity representation $\mathbf{X}$ and the indicator matrices $\mathbf{F}$ and $\mathbf{G}$ iteratively until convergence. The regularized M-FOCUSS algorithm, which solves joint sparse representation, needs a complexity of $O\left(D^3 + 2WD^2 + 4DW^2 + WP^2D + P^2W\right)$, where P denotes the average size of the superpixel patch. The complexity for the indicator matrix $\mathbf{F}$ is $O\left(WNK + 2WK^2 + NK^2\right)$, where K denotes the number of clusters. The complexity for $\mathbf{G}$ is $O\left(NWK + WK^2 + 2NK^2\right)$.

IV. Experimental Results and Discussion

In this section, three typical HSIs are used to assess the effectiveness of the GCSSC method. The following ten benchmark clustering methods are used for comparison: k-means [5], FCM [6], FCM_S1 [42], Ncut [43], USPEC [44], SGCNR [45], RSC [46], SSC [13] (in Algorithm 1), SSSC [17], and S^5C [47].

A. Experimental Setup

1) Comparison Algorithms: To validate the effectiveness of the GCSSC method, the working mechanisms of ten benchmark clustering methods in HSIs are expressed as follows.

***k*-Means [5]:** It iteratively updates the centroids of clusters to find clusters and assigns the pixels to one of the clustering centroids in the feature space.

FCM [6]: It groups pixels into clusters by introducing the fuzziness for the belongingness of image pixels. A pixel belongs to the cluster with a high correlation between pixel and cluster, and far away from the cluster with a low correlation between pixel and cluster.

FCM_S1 [42]: It is an FCM with spatial constraints. It illustrates the fuzziness of the belongingness of each pixel and explores the spatial contextual information of the image.

Ncut [43]: It is a classical spectral clustering method, first learns the affinity matrix, and then applies the centroid-based clustering method for the eigenvector of the Laplacian matrix of the similarity graph.

USPEC [44]: It is scalable and robust for spectral clustering and can be divided into two steps. It constructs a set of base clusters by ultrascalable spectral clustering and then combines multiple base clusters into consensus clustering with more robustness.

SGCNR [45]: It constructs the similarity matrix by an anchor graph and then adds nonnegative relaxation to constrain the cluster indicator matrix.

RSC [46]: It proposes the l_2-norm graph based on the intrasubspace projection dominance of the l_2-norm-based projection space to construct a sparse similarity graph.

SSC [13]: It uses the reconstruction coefficients of the self-representation model to construct the affinity matrix and imposes the l_1 norm on the coefficients.

SSSC [17]: It is an out-of-sample extension for SSC. It first chooses small samples as in-sample data and performs SSC over in-sample data. Then, the out-of-sample data are encoded as a linear combination of the in-sample data and assigned to the cluster.

S^5C [47]: It selects subsamples based on the approximated subgradients to construct the similarity matrix and solves the spectral clustering step by performing eigenvalue decomposition using orthogonal iteration.

In all of the above methods, the number of clusters is set by referencing the ground truth in each dataset. For graph-based clustering methods, such as Ncut, we utilize the self-tuned Gaussian distribution to construct a similarity graph. For nongraph-based clustering methods, such as FCM, FCM_S1, and k-means, the initial selection of centroids is very sensitive; thus, we run repeatedly and report the optimal performance in this article. In all clustering methods, the parameters were manually determined to the optimum.

2) Evaluation Metrics: It is well-known that the evaluation metrics of different clustering methods in HSIs include the overall accuracy (OA), producer's accuracy (PA), and user's accuracy (UA). Specifically, OA is the percentage of the total number of correctly classified pixels to the total number of reference pixels in HSIs. PA is the percentage of correctly classified pixels in each category to the total number of pixels in the corresponding category in the image. UA is the correctly classified pixel in each category divided by the total number of pixels that were assigned in that category. Furthermore, the adjusted mutual information (AMI), normalized mutual information (NMI), adjusted rand index (ARI), Fowlkes–Mallows scores (FMI), V-measure, completeness, homogeneity, and accuracy (ACC), which have different

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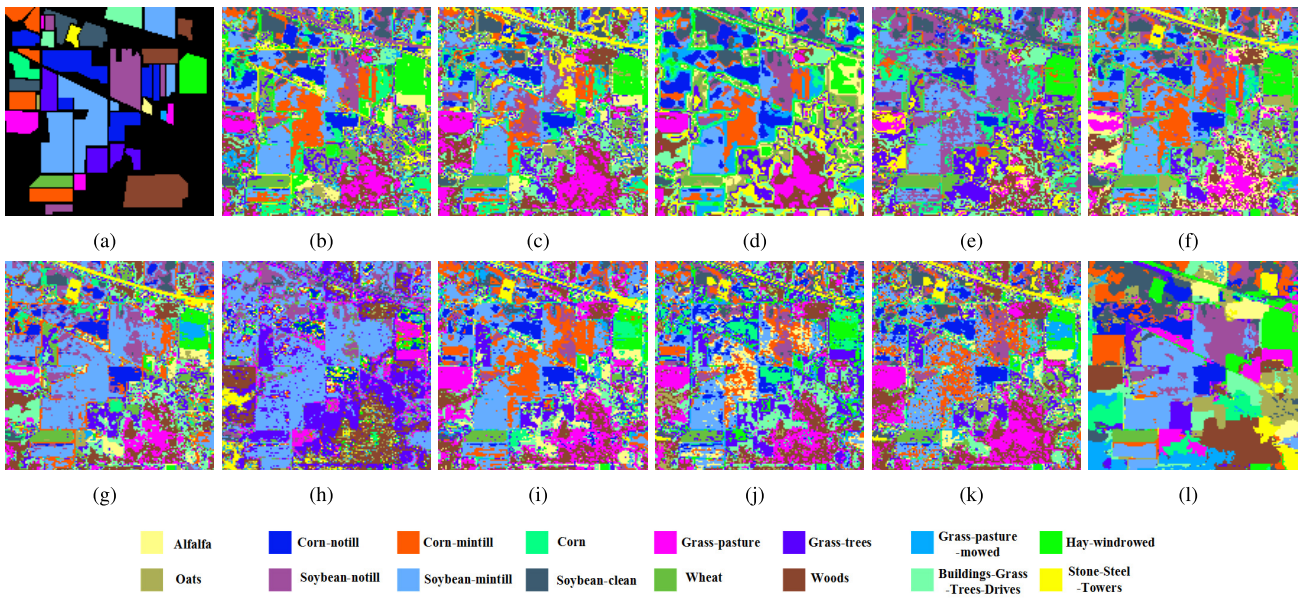


Fig. 3. Clustering results of Indian Pines image. (a) Ground truth. (b) k-means method. (c) FCM method. (d) FCM_S1 method. (e) Ncut method. (f) USPEC method. (g) SGCNR method. (h) RSC method. (i) SSC method. (j) SSSC method. (k) S^5C method. (l) GCSSC method.

working mechanisms, are applied in the HSIs to validate the clustering performance of different methods.

B. Dataset Descriptions and Experimental Analysis

1) Indian Pines Image: The first experimental dataset is the Indian Pines image acquired by AVIRIS spectrograph in 1992, which is farmland in western North Indiana. The Indian Pines image has a size of 145 × 145 with 16 different land-cover classes. To better explore the effectiveness of the experiment, the bands affected by atmospheric absorption and noise are deleted, and the remaining 200 bands are applied to all clustering methods. Maps of the clustering results achieved by different clustering methods are shown in Fig. 3, and the corresponding clustering accuracies are listed in detail in Tables I and II. In Tables I and II, the bold and underlined parts of each row indicate the optimal and second best clustering accuracies, respectively.

From the clustering accuracy obtained by different clustering methods in Table I, we can easily summarize the following results for the Indian Pines image. First, in nongraph-based clustering methods, k-means achieves the best clustering results in OA. The clustering results illustrate that k-means can better distinguish similar and mixed spectral features than FCM and FCM_S1. Specifically, k-means achieves improvements in terms of OA of almost 0.54% and 2.22% over FCM and FCM_S1, respectively. Second, for graph-based clustering methods, SGCNR achieves the best clustering performance by leaning an anchor graph to explore the similarity among hyperspectral pixels. Compared with Ncut, USPEC, RSC, SSC, SSSC, and S^5C, SGCNR has higher clustering accuracy and is improved by 3.77%, 4.62%, 4.73%, 1.79%, 5.38%, and 2.59% in terms of OA, respectively. Specifically, the clustering accuracy in terms of the PA is increased by 18.52%, 18.52%, 7.41%, 11.11%, 16.67%, and 7.41% for the Alfalfa class, respectively; the clustering accuracy in terms of the PA is increased by 70%, 15%, 70%, 70%, 70%, and 50% for the Oats class, respectively. Third, graph-based clustering methods are superior to nongraph-based methods because graph-based clustering methods convert high-dimensional hyperspectral pixels into a graph-cut problem by constructing a similarity among hyperspectral pixels to explore the complex nonlinear structure of HSIs. Specifically, compared with k-means, the SGCNR achieves an improvement in terms of OA of most 4.29%. Fourth, the proposed GCSSC method is more effective and generates a smoother clustering map than the graph-based clustering methods. Compared with Ncut, USPEC, SGCNR, and RSC, the proposed GCSSC obtains superior clustering accuracy and achieves improvements in terms of OA of almost 14.46%, 15.31%, 10.69%, and 15.42%, respectively. Specifically, the clustering accuracy in terms of the PA is increased by 23.36%, 13.46%, 11.93%, and 29.08% for the Corn-notill class, respectively; the clustering accuracy in terms of the PA is increased by 13.09%, 6.55%, 39.27%, and 77.10% for the Hay-windrowed class, respectively; and the clustering accuracy in terms of the PA is increased by 100%, 45%, 30%, and 100% for the Oats class, respectively. Compared with SSC, SSSC, and S^5C, the proposed GCSSC explores the co-occurrence structure between samples and features and achieves improvements in terms of OA of almost 12.48%, 16.07%, and 13.28%, respectively. Specifically, the clustering accuracy in terms of the PA is increased by 14.16%, 20.92%, and 10.11% for the Corn-notill class, respectively; the clustering accuracy in terms of the PA is increased by 41.72%, 45.81%, and 43.77% for the Hay-windrowed class, respectively; and the clustering accuracy in terms of the PA is increased by 100%, 100%, and 80% for the Oats class, respectively. Therefore, the proposed GCSSC method is superior to the other methods for the Indian Pines image both visually and quantitatively and achieves the best clustering results with the best OA of 53.09%.

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TABLE I

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE INDIAN PINES IMAGE [PA, UA, AND OA (%)]

Method	Class	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSC	SSSC	S^5C	GCSSC
Producer's accuracy (PA) (%)	Alfalfa	11.11	20.37	**48.15**	0	0	18.52	11.11	7.41	1.85	11.11	0
	Corn-notill	26.22	26.64	39.54	19.53	29.43	30.96	13.81	28.73	21.97	32.78	**42.89**
	Corn-mintill	35.37	27.10	26.86	0.48	40.89	3.12	4.20	**44.36**	34.65	32.61	43.17
	Corn	41.45	28.21	36.32	**61.11**	19.23	20.09	8.55	2.56	32.48	8.97	0.43
	Grass-pasture	48.69	58.15	53.72	31.79	39.24	44.87	18.11	62.78	60.36	**62.78**	27.77
	Grass-trees	28.92	28.25	27.98	42.17	30.39	45.25	**74.56**	67.34	55.15	59.84	55.69
	Grass-pasture-mowed	0	65.38	0	0	0	0	0	69.23	50.00	**76.92**	0
	Hay-windrowed	98.36	93.05	60.33	86.71	93.25	60.53	22.70	58.08	53.99	56.03	**99.80**
	Oats	0	10.00	35.00	0	55.00	70.00	0	0	0	20.00	**100**
	Soybean-notill	38.84	29.96	42.67	**55.06**	31.92	33.26	22.83	21.07	27.07	28.51	52.38
	Soybean-mintill	38.01	36.87	30.15	49.96	40.44	67.46	**71.76**	47.08	36.02	41.69	65.28
	Soybean-clean	29.97	23.62	31.60	16.94	18.73	13.19	7.17	9.28	11.40	18.24	**44.30**
	Wheat	90.57	95.28	56.60	99.53	95.75	95.28	7.55	93.40	77.83	83.49	**99.53**
	Woods	37.64	44.67	36.86	40.03	33.38	42.66	**63.52**	40.42	47.76	46.14	57.65
	Buildings-Grass-Trees-Drives	15.79	18.42	24.21	21.32	20.79	23.95	3.42	18.95	26.05	8.16	**31.05**
	Stone-Steel-Towers	0	42.11	0	0	87.37	88.42	1.05	**89.47**	67.37	83.16	0
User's accuracy (UA) (%)	Alfalfa	5.94	5.67	6.34	0	0	**14.29**	2.75	1.97	0.12	3.61	0
	Corn-notill	45.63	52.26	44.26	**64.81**	56.27	56.63	47.48	53.51	37.37	49.84	61.07
	Corn-mintill	18.19	21.18	17.81	1.79	18.58	5.37	11.25	16.09	21.86	15.64	**40.31**
	Corn	12.37	10.58	21.09	21.70	11.60	5.77	**23.81**	1.83	10.26	9.21	11.11
	Grass-pasture	34.47	29.46	33.93	44.38	30.09	31.81	23.68	31.11	34.68	30.32	**78.41**
	Grass-trees	69.01	76.45	80.08	61.17	82.25	81.06	31.38	78.23	57.54	76.28	**97.42**
	Grass-pasture-mowed	0	**10.43**	0	0	0	0	0	5.10	8.02	7.38	0
	Hay-windrowed	62.39	89.22	57.06	**91.77**	88.72	87.57	35.46	86.85	54.21	82.53	74.96
	Oats	0	0.64	2.15	0	2.86	**8.00**	0	0	0	2.19	4.34
	Soybean-notill	**56.54**	25.28	52.68	19.60	35.44	22.56	24.42	39.16	34.25	34.72	38.53
	Soybean-mintill	47.33	51.41	59.66	54.25	48.90	49.09	43.25	46.99	52.02	48.47	**69.08**
	Soybean-clean	25.92	27.88	25.53	23.27	25.61	27.27	**39.29**	22.89	22.29	24.14	27.98
	Wheat	32.76	42.89	29.56	25.54	33.50	40.81	6.48	67.12	70.82	49.30	**100**
	Woods	85.59	80.06	91.73	**93.67**	85.21	88.32	69.96	88.49	84.89	84.08	69.59
	Buildings-Grass-Trees-Drives	20.07	28.69	25.84	16.56	34.05	17.17	28.89	35.12	23.63	19.62	**71.08**
	Stone-Steel-Towers	0	6.32	0	0	**91.21**	82.35	0.98	82.52	37.87	28.21	0
OA		38.11	37.57	35.89	38.63	37.78	42.40	37.67	40.61	37.02	39.81	**53.09**

TABLE II

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE INDIAN PINES IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSC	SSSC	S^5C	GCSSC
NMI	43.86	43.80	42.17	43.74	44.70	46.36	29.72	45.75	39.03	43.05	**56.38**
AMI	42.17	41.33	39.43	42.99	42.87	45.83	27.18	42.48	38.75	42.78	**56.17**
ARI	21.95	22.95	20.70	22.12	22.48	26.45	20.14	23.18	20.24	23.54	**36.51**
FMI	30.71	31.25	29.02	32.47	31.28	36.64	33.76	33.11	28.83	32.40	**44.00**
V-measure	43.84	43.74	42.10	43.74	44.67	46.35	29.64	42.75	39.03	43.05	**56.38**
Completeness	42.43	41.59	39.69	44.23	43.12	46.62	32.10	42.95	37.37	41.93	**55.51**
Homogeneity	45.35	46.13	44.81	43.26	46.34	46.09	27.53	42.55	40.85	44.24	**57.28**
ACC	38.11	37.57	35.89	38.63	37.78	42.40	37.67	40.61	37.02	39.81	**53.09**

To further validate the effectiveness of GCSSC, the clustering results of different evaluation metrics (NMI, AMI, ARI, FMI, V-measure, completeness, homogeneity, and ACC) are listed in Table II for the Indian Pines image. From Table II, we clearly observe that the proposed GCSSC obtains better clustering performance than the other clustering methods. Specifically, in the nongraph-based clustering methods, k-means obtains the best clustering result; however, FCM_S1 achieves poor clustering accuracy and is not well separated. In the graph-based clustering methods, SCGNR and SSC achieve better clustering accuracy. Compared with SCGNR and SSC, the proposed GCSSC obtains the best clustering performance based on the visual effects and accuracies. This further illustrates that the proposed GCSSC method can exploit the intrinsic structure characteristics and boost the clustering performance simultaneously for the Indian Pines dataset.

2) Salinas Image: The second experimental dataset is the Salinas image acquired by the AVIRIS spectrograph in 1998. This image is of a valley in California. The Salinas image has a size of 512 $\times$ 217 with 16 different land-cover classes. We remove the bands affected by water absorption and noise and retain 204 bands for all clustering methods. Maps of the clustering results achieved by the different clustering methods are expressed in Fig. 4, and the corresponding clustering accuracies are given in detail in Tables III and IV.

The clustering accuracies and clustering maps of different clustering methods are shown in Table III and Fig. 4 for the Salinas image, respectively. From Table III, we can easily summarize the following results. First, for nongraph-based clustering methods, k-means achieves better clustering accuracy and successfully identifies more categories than FCM and FCM_S1; however, FCM_S1 achieves smoother maps

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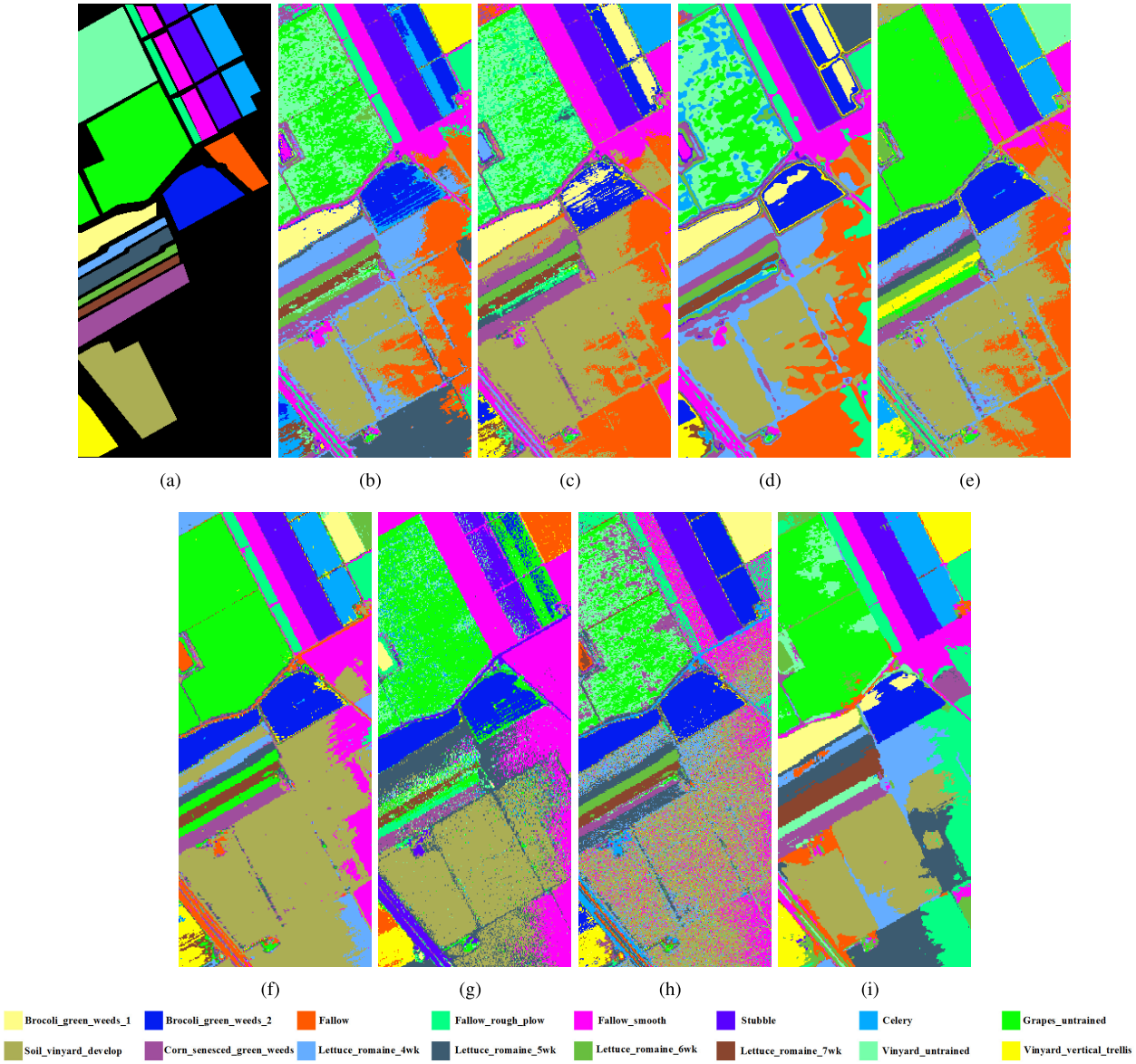


Fig. 4. Clustering results of Salinas image. (a) Ground truth. (b) k-means method. (c) FCM method. (d) FCM_S1 method. (e) USPEC method. (f) SGCNR method. (g) SSSC method. (h) S^5C method. (i) GCSSC method.

than k-means. Second, for the graph-based clustering methods, SGCNR achieves more effective performance and improvement in terms of OA of almost 0.34%, 12.31%, and 14.03% in contrast to USPEC, SSSC, and S^5C, respectively. Specifically, from the perspective of the Fallow_rough_plow class, the PA increased by 99.35%, 99.35%, and 15.99%, respectively, and from the perspective of the Lettuce_romaine_7wk class, the PA increased by 92.90%, 29.82%, and 3.83%, respectively. Third, for the quantitative evaluation acquired by nongraph- and graph-based clustering methods, graph-based clustering methods improve the clustering accuracy by reducing most of the misclassifications. Specifically, compared with k-means, the clustering accuracy of SGCNR in terms of the OA increased by 1.42%. Fourth, compared with nongraph-based methods, the proposed GCSSC achieves significant improvement. Specifically, the clustering performance of GCSSC in terms of OA is improved by 8.67%, 19.93%, and 13.48% compared to k-means, FCM, and FCM_S1, respectively. Fifth, from the perspective of OA, the proposed GCSSC achieves the optimal clustering performance and obtains improvements of almost 7.59%, 7.25%, 19.56%, and 21.28% compared with USPEC, SGCNR, SSSC, and S^5C, respectively. Specifically, the clustering accuracy in terms of the PA is increased by 100%, 100%, 100%, and 100% for the Brocoli_green_weeds_1 class, respectively; the clustering accuracy in terms of the PA is increased by 49.87%, 46.97%, 23.56%, and 28.03% for the Lettuce_romaine_5wk class, respectively; and the clustering accuracy in terms of the PA is increased by 21.75%, 16.60%, 11.51%, and 17.71% for the Vinyard_vertical_trellis class, respectively. Therefore, the GCSSC proposed in this article is superior to the other typical methods in HSI clustering both visually and quantitatively and achieves the best clustering accuracy with an OA of 76.66%.

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TABLE III

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE SALINAS IMAGE [PA, UA, AND OA (%)]. ("-" DENOTES THE OUT OF MEMORY)

Method	Class	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSC	SSSC	S^5C	GCSSC
Producer's	Brocoli_green_weeds_1	98.51	99.80	99.05	-	0	0	-	-	0	0	**100**
accuracy	Brocoli_green_weeds_2	83.90	70.29	81.45	-	**98.42**	96.22	-	-	87.98	97.88	85.53
(PA)	Fallow	38.01	15.79	33.60	-	**42.00**	0.10	-	-	0	4.25	0
(%)	Fallow_rough_plow	99.71	0	99.07	-	0	99.35	-	-	0	83.36	**99.86**
	Fallow_smooth	99.22	99.10	93.61	-	96.71	99.10	-	-	**99.89**	76.89	99.25
	Stubble	99.44	99.01	96.74	-	99.39	99.39	-	-	80.90	99.47	**99.77**
	Celery	38.53	0	0.03	-	99.36	97.46	-	-	0.36	0.03	**99.61**
	Grapes_untrained	58.72	42.09	50.94	-	97.18	**98.94**	-	-	82.97	52.86	92.56
	Soil_vinyard_develop	94.44	98.15	92.28	-	96.07	**98.34**	-	-	96.76	64.24	89.38
	Corn_senesced_green_weeds	56.35	63.27	45.79	-	59.46	63.79	-	-	45.15	24.67	**64.70**
	Lettuce_romaine_4wk	96.82	0	**99.91**	-	98.31	3.00	-	-	0	56.46	66.39
	Lettuce_romaine_5wk	0.16	0.21	0	-	45.36	48.26	-	-	71.67	67.20	**95.23**
	Lettuce_romaine_6wk	98.03	97.05	95.85	-	97.49	0	-	-	18.45	**98.14**	0
	Lettuce_romaine_7wk	79.53	81.50	88.79	-	0	92.90	-	-	63.08	89.07	**98.60**
	Vinyard_untrained	**61.53**	51.39	54.64	-	0	0	-	-	18.82	46.62	20.83
	Vinyard_vertical_trellis	0	43.88	53.02	-	62.48	67.63	-	-	72.72	66.52	**84.23**
User's	Brocoli_green_weeds_1	**99.85**	41.79	48.60	-	0	0	-	-	0	0	78.78
accuracy	Brocoli_green_weeds_2	56.98	55.24	56.37	-	64.65	64.02	-	-	52.55	38.26	**100**
(UA)	Fallow	97.41	85.01	**100**	-	89.25	33.33	-	-	0	16.00	0
(%)	Fallow_rough_plow	98.65	0	95.37	-	0	**99.50**	-	-	0	72.90	89.69
	Fallow_smooth	**78.56**	55.40	73.65	-	59.88	68.30	-	-	41.61	55.13	70.52
	Stubble	99.52	99.90	97.04	-	99.95	99.87	-	-	**100**	99.75	98.75
	Celery	48.10	0	0.03	-	86.48	97.98	-	-	0.73	1.09	**99.92**
	Grapes_untrained	73.09	77.72	**84.12**	-	58.52	54.78	-	-	51.27	73.89	64.77
	Soil_vinyard_develop	97.50	69.69	96.02	-	87.25	74.96	-	-	96.11	70.69	**99.55**
	Corn_senesced_green_weeds	61.69	55.07	54.29	-	78.05	89.59	-	-	**95.79**	20.11	71.66
	Lettuce_romaine_4wk	33.70	0	30.13	-	50.90	3.62	-	-	0	24.17	**84.71**
	Lettuce_romaine_5wk	2.73	0.47	0	-	84.61	83.11	-	-	45.94	36.94	**87.72**
	Lettuce_romaine_6wk	35.48	81.34	46.21	-	48.48	0	-	-	**86.22**	82.63	0
	Lettuce_romaine_7wk	55.84	65.12	74.69	-	0	66.49	-	-	**94.27**	84.64	50.62
	Vinyard_untrained	49.94	42.79	**50.64**	-	0	0	-	-	36.03	46.32	50.17
	Vinyard_vertical_trellis	0	83.21	84.33	-	51.98	86.30	-	-	99.02	85.49	**99.54**
OA		67.99	56.73	63.18	-	69.07	69.41	-	-	57.10	55.38	**76.66**

TABLE IV

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE SALINAS IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSC	SSSC	S^5C	GCSSC
NMI	73.56	68.59	70.89	-	78.91	79.86	-	-	66.20	64.84	**83.11**
AMI	72.62	66.80	70.56	-	72.29	71.57	-	-	66.17	64.81	**83.10**
ARI	54.66	46.93	51.19	-	63.43	59.22	-	-	45.53	43.21	**67.22**
FMI	59.28	52.42	55.96	-	70.39	68.06	-	-	54.13	49.14	**71.53**
V-measure	73.56	68.56	70.89	-	78.61	79.38	-	-	66.20	64.84	**83.11**
Completeness	74.50	70.39	70.59	-	86.12	**89.08**	-	-	72.06	66.28	84.95
Homogeneity	72.64	66.83	71.20	-	72.31	71.59	-	-	61.22	63.46	**81.35**
ACC	67.99	56.73	63.18	-	69.07	69.41	-	-	57.10	55.38	**76.66**

In addition, quantitative evaluations with different working mechanisms, such as NMI, AMI, ARI, FMI, V-measure, completeness, homogeneity, and ACC, are shown in Table IV for the Salinas image. From Table IV, the GCSSC is further verified to achieve the optimal clustering performance compared with other state-of-the-art clustering methods. Specifically, from the perspective of nongraph-based clustering methods, k-means in all evaluation metrics achieves the best clustering performance in contrast to FCM and FCM_S1. Compared with nongraph-based methods, USPEC and SGCNR successfully identify most land-cover classes to boost the clustering performance and obtain better clustering results. Through comparison and analysis of USPEC, SGCNR, and GCSSC in terms of NMI, AMI, ARI, FMI, V-measure, completeness, homogeneity, and ACC, the proposed GCSSC achieves remarkable clustering performance. In conclusion, these evaluation performances further illustrate the superiority of GCSSC.

3) Pavia Center Image: The third experimental dataset is the Pavia Center image acquired by the ROSIS sensor in 2003. The image is of the city of Pavia, Italy. The spectral covering range is 0.43–0.86 μm with 115 spectral bands. However, in the Pavia Center image, some bands with high noise and some samples with no information are discarded and removed in this experiment. In this article, we used a test image with a size of 1096 $\times$ 490 consisting of nine different land-cover classes and 102 spectral bands. Maps of the clustering results achieved by the different clustering methods are shown in Fig. 5, and the corresponding clustering accuracies are listed in detail in Tables V and VI.

Table V and Fig. 5 illustrate that the proposed GCSSC method is more effective than the other clustering methods. In the nongraph-based clustering results, FCM_S1 obtains better clustering results and smoother maps, and the improvement is increased by 0.95% and 5.33% compared to k-means

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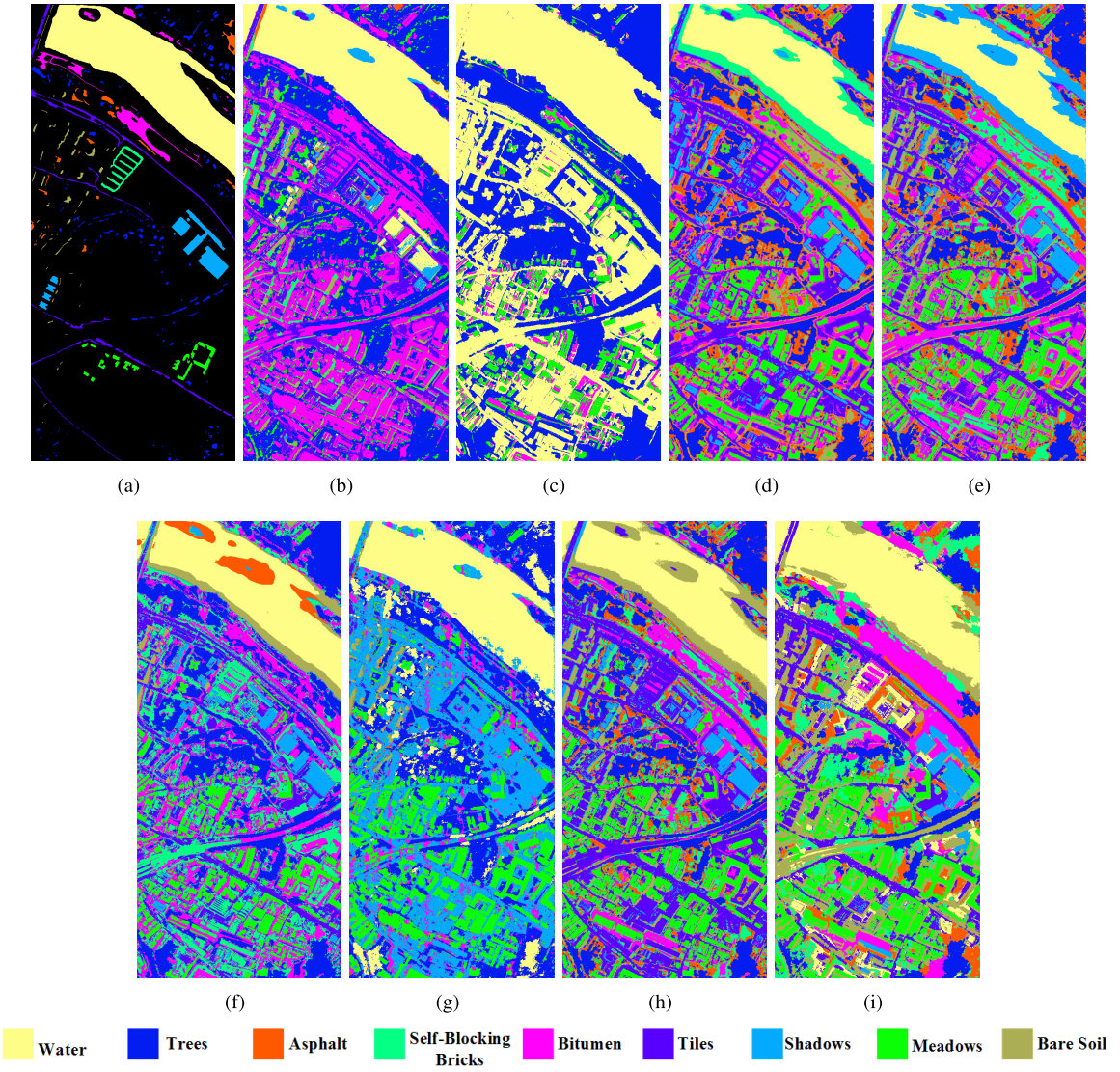


Fig. 5. Clustering results of Pavia Center image. (a) Ground truth. (b) k-means method. (c) FCM method. (d) FCM_S1 method. (e) USPEC method. (f) SGCNR method. (g) SSSC method. (h) S^{5}C method. (i) GCSSC method.

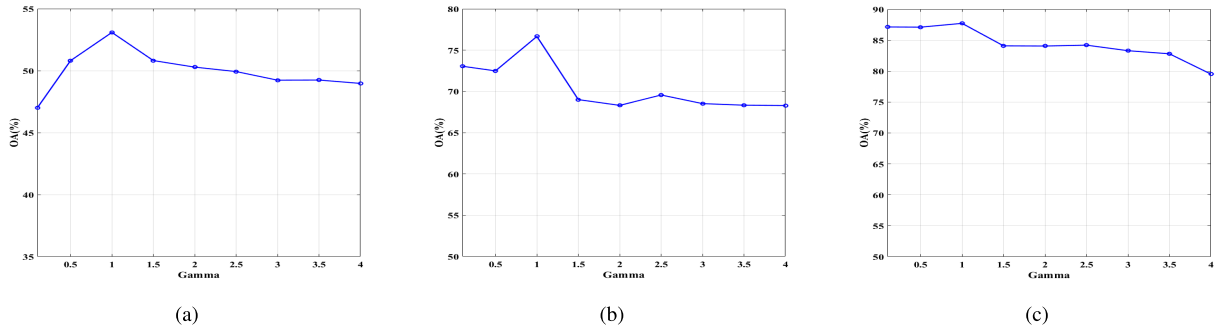


Fig. 6. Parameter analysis of γ. (a) Indian Pines image. (b) Salinas image. (c) Pavia center image.

and FCM, respectively. For the graph-based clustering methods, S^{5}C obtains an improvement in OA of almost 4.36%, 3.09%, and 2.83% compared to USPEC, SGCNR, and SSSC, respectively. Compared with k-means, FCM, and FCM_S1, the GCSSC significantly decreases most of the misclassifications by improving by 10.53%, 14.91%, and 9.58% in terms of OA, respectively. Compared with USPEC, SGCNR, SSSC, and S^{5}C, the GCSSC obtains improvements of 10.41%, 9.14%, 8.88%, and 6.05% in terms of OA, respectively.

For the Pavia Center image, the different evaluation metrics are shown in Table VI to further illustrate the clustering performance. In Table VI, the proposed GCSSC achieves the

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TABLE V

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE PAVIA CENTER IMAGE [PA, UA, AND OA (%)]

Method	Class	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSC	SSSC	S^5C	GCSSC
Producer's	Water	97.50	**100**	82.99	-	77.19	83.23	-	-	96.08	85.13	94.83
accuracy	Trees	99.97	**100**	97.23	-	89.33	98.20	-	-	65.74	90.01	64.98
(PA)	Asphalt	0	0	34.45	-	55.03	0	-	-	0	50.16	**66.00**
(%)	Self-Blocking Bricks	0.05	0	0	-	0	**77.66**	-	-	0	0	0
	Bitumen	64.64	18.08	45.40	-	58.39	46.95	-	-	36.01	58.88	**73.81**
	Tiles	43.77	0	89.00	-	97.89	84.00	-	-	11.56	**99.40**	84.74
	Shadows	17.32	0	85.92	-	83.60	84.51	-	-	**99.30**	84.27	86.66
	Meadows	0	75.40	99.33	-	98.33	91.96	-	-	**99.97**	99.90	99.78
	Bare Soil	42.40	0.32	14.64	-	85.22	22.73	-	-	45.40	47.39	**95.94**
User's	Water	92.85	76.61	100	-	100	100	-	-	96.01	**100**	95.31
accuracy	Trees	65.79	52.94	71.32	-	78.56	60.49	-	-	55.30	79.86	**98.60**
(UA)	Asphalt	0	0	57.21	-	68.90	0	-	-	0	69.15	**79.93**
(%)	Self-Blocking Bricks	0.20	0	0	-	0	**27.43**	-	-	0	0	0
	Bitumen	36.25	61.16	89.68	-	82.17	75.48	-	-	72.91	**99.95**	97.76
	Tiles	43.61	0	59.77	-	68.83	86.19	-	-	**89.51**	55.59	78.33
	Shadows	41.03	0	79.49	-	28.97	89.82	-	-	35.04	96.41	**99.40**
	Meadows	0	58.53	70.88	-	99.97	**100**	-	-	69.63	97.93	99.84
	Bare Soil	74.57	**100**	25.65	-	99.57	14.41	-	-	96.47	9.63	32.07
OA		77.19	72.81	78.14	-	77.31	78.58	-	-	78.84	81.67	**87.72**

TABLE VI

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE PAVIA CENTER IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSC	SSSC	S^5C	GCSSC
NMI	67.53	54.19	72.63	-	72.43	71.09	-	-	64.71	76.06	**76.62**
AMI	61.96	36.10	67.83	-	67.85	65.26	-	-	64.71	76.05	**76.61**
ARI	79.14	46.91	71.70	-	63.51	71.49	-	-	79.30	74.48	**82.11**
FMI	88.49	77.04	82.71	-	77.32	82.58	-	-	88.30	84.49	**89.56**
V-measure	67.28	50.01	72.46	-	72.28	70.83	-	-	64.71	76.06	**76.62**
Completeness	73.58	**81.31**	67.83	-	67.86	65.27	-	-	71.52	72.10	76.04
Homogeneity	61.97	36.11	77.77	-	77.31	77.44	-	-	59.09	**80.47**	77.21
ACC	77.19	72.81	78.14	-	77.31	78.58	-	-	78.84	81.67	**87.72**

TABLE VII

QUANTITATIVE EVALUATION IN TERMS OF OA WITH AND WITHOUT GC

Data sets	Without GC	With GC
Indian Pines	49.59	53.09
Salinas	66.70	76.66
Pavia Center	78.05	87.72

best evaluation performance compared to the other clustering methods. In conclusion, the evaluation results in Table VI further illustrate the effectiveness of GCSSC.

C. Effectiveness of Graph Convolution in Dictionary Atoms

To validate the effectiveness of the graph convolution constraint (GC) for dictionary atoms, we compare the clustering accuracy in terms of OA with and without the spectral graph convolution operation. The results are given in Table VII in detail. The graph convolution operation is incorporated into the dictionary atoms to capture the spectral similarity and geometric structure relationship between dictionary atoms. Table VII indicates that the method with GC is more effective than that without GC for all three HSIs.

D. Parameter Analysis

In this section, we study four important parameters, parameters γ, α, W, and $\mathcal{N}(\tilde{\mathbf{y}}_j)$, which have some influence on the clustering performance. Leave-one-out cross-validation is utilized to analyze the sensitivity of each parameter in detail. Figs. 6–9 show the changes curve in clustering accuracy OA with respect to each parameter on three experimental images.

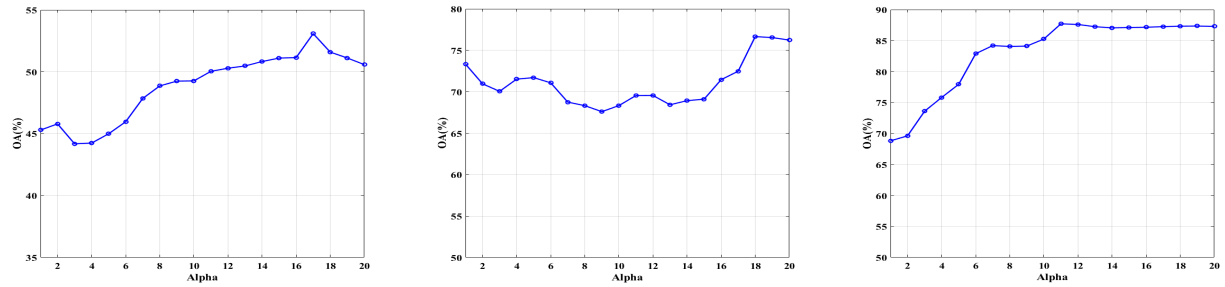
1) Parameter γ: Parameter γ is utilized to balance the clustering error and representation accuracy. The range of parameter γ for each of the three HSIs is set from 0.1 to 4. From Fig. 6, we clearly observe that the best performance with the highest OA is obtained when parameter γ is set to 1 for all three hyperspectral datasets.

2) Parameter α: Parameter α denotes the tradeoff parameter between the reconstruction error and sparsity regularization. The range of the parameter α for each of the three HSIs is set from 1 to 20. From Fig. 7, we can clearly observe that parameter α has a stable influence on the clustering accuracy. For the Indian Pines dataset, the Salinas dataset, and the Pavia Center dataset, the best clustering results in terms of OAs are shown when parameter α is set to 17, 18, and 11, respectively.

3) Parameter W: Parameter W denotes the number of dictionary atoms, which is obtained by performing the superpixel segmentation. Parameter W is utilized to control the representation ability and the redundancy of dictionary atoms. The range of parameter W for each of the three HSIs is set from 50 to 800. From Fig. 8, it can be observed that parameter W is very important for the proposed GCSSC method, and it achieves the optimal clustering accuracy with the highest OA when parameter W is set to 100, 400, and 600 for Indian Pines images, Salinas image, and Pavia Center image, respectively.

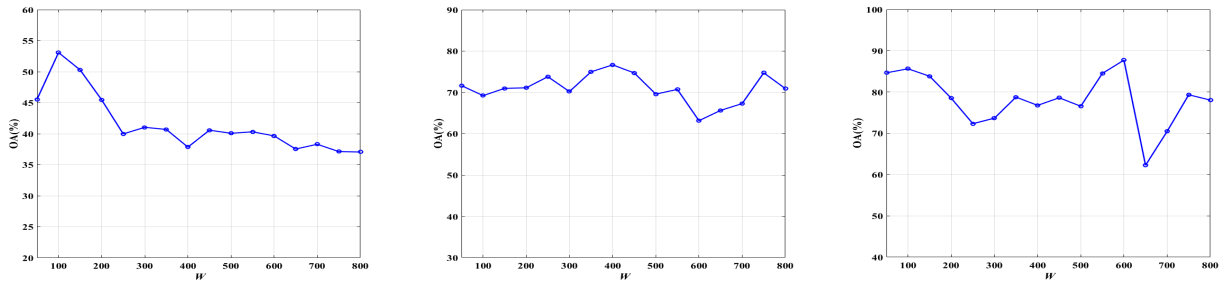
4) Parameter $\mathcal{N}(\tilde{\mathbf{y}}_j)$: Parameter $\mathcal{N}(\tilde{\mathbf{y}}_j)$ denotes the number of spectral nearest neighbors of dictionary atoms, which relates

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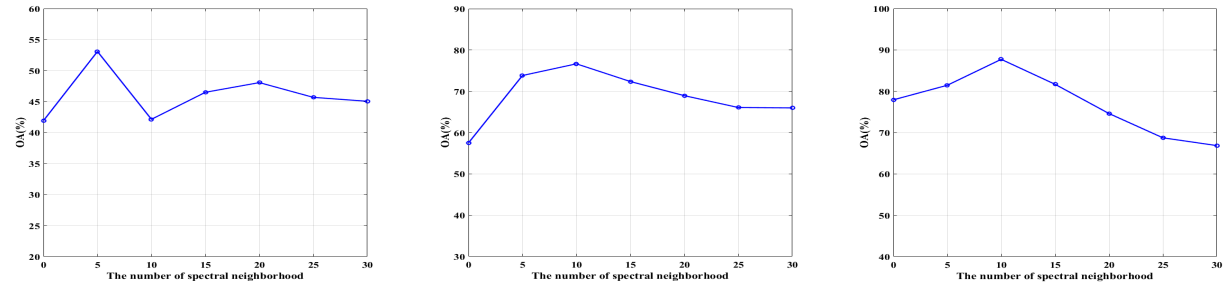
(a) (b) (c)

Fig. 7. Parameter analysis of α. (a) Indian Pines image. (b) Salinas image. (c) Pavia center image.



(a) (b) (c)

Fig. 8. Parameter analysis of W. (a) Indian Pines image. (b) Salinas image. (c) Pavia center image.



(a) (b) (c)

Fig. 9. Parameter analysis of $\mathcal{N}\left(\tilde{\mathbf{y}}_j\right)$. (a) Indian Pines image. (b) Salinas image. (c) Pavia center image.

TABLE VIII
EXECUTION TIME OF DIFFERENT ALGORITHMS (SECONDS)

Data sets	k-means	FCM	FCM_S1	Ncut	USPEC	SGCNR	RSC	SSC	SSSC	S^5C	GCSSC
Indian Pines	8.51	127.82	163.32	25.81	2.23	62.32	2411.37	1181.38	5.24	307.01	194.72
Salinas	23.87	990.82	967.36	-	4.07	91.69	-	-	40.46	1343.40	10681.04
Pavia Center	73.14	79.03	1684.03	-	15.10	707.66	-	-	122.91	5298.25	12238.92

the spectral distribution of HSI. It is determined by the similarity of the spectral neighborhood for HSIs. The range of parameter $\mathcal{N}(\tilde{\mathbf{y}}_j)$ for each of the three HSIs is set from 0 to 30. As shown in Fig. 9, the influence of clustering accuracy is relatively stable for various values of $\mathcal{N}(\tilde{\mathbf{y}}_j)$, and the good performance with the highest OA is obtained when parameter $\mathcal{N}(\tilde{\mathbf{y}}_j)$ is set to 5, 10, and 10 for Indian Pines images, Salinas image, and Pavia Center image, respectively.

E. Computational Time

In this section, the average computation times of different clustering methods are described in Table VIII. The experimental environment for all methods is given as follows: i7-8700 K CPU @ 3.70 GHz, 3.70 GHz, 32.00-GB RAM, and MATLAB R2016b.

In Table VIII, USPEC requires the shortest running time in all clustering methods. Compared with the nongraph-based clustering methods, graph-based clustering methods require more running time because they need to construct the graph among hyperspectral pixels. Although the proposed GCSSC is slower than FCM, FCM_S1, k-means, Ncut, USPEC, SGCNR, SSSC, and S^5C, it obtains better clustering accuracy. Compared with RSC and SSC, the proposed GCSSC requires a shorter running time.

V. CONCLUSION

In this article, we proposed a novel GCSSC algorithm for large HSIs. The key to the GCSSC is to integrate spectral

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coclustering and sparse subspace learning into a unified framework and iteratively update the low-rank indicator matrix and affinity matrix until optimal clustering accuracies are achieved. To better explore the intrinsic affinity matrix in HSIs, we considered graph convolution to aggregate the local neighborhood information between dictionary atoms and further improve the representation ability for HSIs. The experimental results demonstrated that the GCSSC is efficient and effective in dealing with large HSIs.

Currently, the proposed GCSSC method only relates to the linearity assumption of the self-representation and has no considering the complexity and nonlinearity of HSI. Motivated by the exceptional ability of deep neural networks to capture complex nonlinear subspaces of data and learn discriminative features for clustering, the future work will focus on extending the subspace clustering to the deep neural networks under the self-supervised framework for large HSIs.

REFERENCES

[1] P. A. Townsend and S. J. Walsh, "Remote sensing of forested wetlands: Application of multitemporal and multispectral satellite imagery to determine plant community composition and structure in southeastern USA," *Plant Ecol.*, vol. 157, no. 2, pp. 129–149, 2001.

[2] R. J. Ellis and P. W. Scott, "Evaluation of hyperspectral remote sensing as a means of environmental monitoring in the St. Austell China clay (kaolin) region, Cornwall, UK," *Remote Sens. Environ.*, vol. 93, nos. 1–2, pp. 118–130, 2004.

[3] A. Mancini, E. Frontoni, and P. Zingaretti, "A multi/hyper-spectral imaging system for land use/land cover using unmanned aerial systems," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, 2016, pp. 1148–1155.

[4] X. Zhu, S. Zhang, Y. Li, J. Zhang, L. Yang, and Y. Fang, "Low-rank sparse subspace for spectral clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 8, pp. 1532–1543, Aug. 2019.

[5] T. Kanungo, D. M. Mount, N. S. Netanyahu, C. D. Piatko, R. Silverman, and A. Y. Wu, "An efficient k-means clustering algorithm: Analysis and implementation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 881–892, Jul. 2002.

[6] T. Yang, C. Lee, and S. Yen, "Fuzzy objective functions for robust pattern recognition," in *Proc. IEEE Int. Conf. Fuzzy Syst.*, Aug. 2009, pp. 2057–2062.

[7] Z. Wu, W. Xie, and J. Yu, "Kernel method-based fuzzy clustering algorithm," *J. Syst. Eng. Electron.*, vol. 16, no. 1, pp. 160–166, Mar. 2005.

[8] W. Zhang, X. Zhang, J. Zhao, Y. Qiang, Q. Tian, and X. Tang, "A segmentation method for lung nodule image sequences based on superpixels and density-based spatial clustering of applications with noise," *PLoS ONE*, vol. 12, no. 9, Sep. 2017, Art. no. e0184290.

[9] M. Ankerst, M. M. Breunig, H.-P. Kriegel, and J. Sander, "OPTICS: Ordering points to identify the clustering structure," *ACM SIGMOD. Rec.*, vol. 28, no. 2, pp. 49–60, Jun. 1999.

[10] I. Davidson and S. S. Ravi, "Agglomerative hierarchical clustering with constraints: Theoretical and empirical results," in *Proc. 9th Eur. Conf. Princ. Pract. Knowl. Discovery Databases*, Oct. 2005, pp. 59–70.

[11] T. Xiong, S. Wang, A. Mayers, and E. Monga, "DHCC: Divisive hierarchical clustering of categorical data," *Data Mining Knowl. Discovery*, vol. 24, no. 1, pp. 103–135, Jan. 2012.

[12] J. Han, K. Xiong, F. Nie, and X. Li, "Structured graph reconstruction for scalable clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 33, no. 5, pp. 2252–2265, May 2021.

[13] E. Elhamifar and R. Vidal, "Sparse subspace clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 2790–2797.

[14] E. Elhamifar and R. Vidal, "Sparse subspace clustering: Algorithm, theory, and applications," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2765–2781, Nov. 2013.

[15] H. Zhai, H. Zhang, L. Zhang, and P. Li, "Total variation regularized collaborative representation clustering with a locally adaptive dictionary for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 166–180, Jan. 2019.

[16] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.

[17] X. Peng, L. Zhang, and Z. Yi, "Scalable sparse subspace clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 430–437.

[18] P. A. Traganitis and G. B. Giannakis, "Sketched subspace clustering," *IEEE Trans. Signal Process.*, vol. 66, no. 7, pp. 1663–1675, Apr. 2018.

[19] H. Zhai, H. Zhang, X. Xu, L. Zhang, and P. Li, "Kernel sparse subspace clustering with a spatial max pooling operation for hyperspectral remote sensing data interpretation," *Remote Sens.*, vol. 9, no. 4, p. 335, 2017.

[20] Y. Cai, Z. Zhang, Z. Cai, X. Liu, X. Jiang, and Q. Yan, "Graph convolutional subspace clustering: A robust subspace clustering framework for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4191–4202, May 2021.

[21] J. Song, G. Yoon, K. Hahn, and S. M. Yoon, "Subspace clustering via structure-enforced dictionary learning," *Neurocomputing*, vol. 362, no. 14, pp. 1–10, Oct. 2019.

[22] Y. Zhang, D. Shi, J. Gao, and D. Cheng, "Low-rank-sparse subspace representation for robust regression," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2972–2981.

[23] S. Huang, Z. Xu, and J. Lv, "Adaptive local structure learning for document co-clustering," *Knowl.-Based Syst.*, vol. 148, pp. 74–84, May 2018.

[24] N. Huang, L. Xiao, and Y. Xu, "Bipartite graph partition based coclustering with joint sparsity for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4698–4711, Dec. 2019.

[25] F. Nie, X. Wang, C. Deng, and H. Huang, "Learning a structured optimal bipartite graph for co-clustering," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 4129–4138.

[26] Y. Zhao, Y. Yuan, F. Nie, and Q. Wang, "Spectral clustering based on iterative optimization for large-scale and high-dimensional data," *Neurocomputing*, vol. 318, no. 27, pp. 227–235, Nov. 2018.

[27] D. Cai, X. He, J. Han, and T. S. Huang, "Graph regularized nonnegative matrix factorization for data representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 8, pp. 1548–1560, Aug. 2011.

[28] C. Ding, X. He, and H. D. Simon, "On the equivalence of nonnegative matrix factorization and spectral clustering," in *Proc. SIAM Int. Conf. Data Mining*, Apr. 2005, pp. 606–610.

[29] L. Zhang, L. Zhang, B. Du, J. You, and D. Tao, "Hyperspectral image unsupervised classification by robust manifold matrix factorization," *Inf. Sci.*, vol. 485, pp. 154–169, Jun. 2019.

[30] F. Nie, D. Chris, D. Luo, and H. Huang, "Improved minmax cut graph clustering with nonnegative relaxation," in *Proc. Mach. Learn. Knowl. Discovery Databases*, 2010, pp. 451–466.

[31] Y. Chen, C.-G. Li, and C. You, "Stochastic sparse subspace clustering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 4154–4163.

[32] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst.*, 2016, pp. 3844–3852.

[33] D. K. Hammond, P. Vandergheynst, and R. Gribonval, "Wavelets on graphs via spectral graph theory," *Appl. Comput. Harmon. Anal.*, vol. 30, no. 2, pp. 129–150, Mar. 2011.

[34] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent.*, 2017, pp. 1–14.

[35] H. Zhai, H. Zhang, L. Zhang, P. Li, and A. Plaza, "A new sparse subspace clustering algorithm for hyperspectral remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 1, pp. 43–47, Jan. 2017.

[36] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. CVPR*, Jun. 2011, pp. 2097–2104.

[37] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher, "An analysis of approximations for maximizing submodular set functions—I," *Math. Programmin*, vol. 14, no. 1, pp. 265–294, 1978.

[38] F. Wang, C. Ding, and T. Li, "Integrated KL (K-means–Laplacian) clustering: A new clustering approach by combining attribute data and pairwise relations," in *Proc. SIAM Int. Conf. Data Mining*, Apr. 2009, pp. 38–48.

[39] Y. Zhang *et al.*, "Unsupervised nonnegative adaptive feature extraction for data representation," *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 12, pp. 2423–2440, Dec. 2019.

[40] A. Soltani-Farani, H. R. Rabiee, and S. A. Hosseini, "Spatial-aware dictionary learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 527–541, Jan. 2015.

[41] D. P. Bertsekas, "Nonlinear programming," *J. Oper. Res. Soc.*, vol. 48, no. 3, p. 334, Jan. 1997.

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[42] S. Chen and D. Zhang, "Robust image segmentation using FCM with spatial constraints based on new kernel-induced distance measure," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 34, no. 4, pp. 1907–1916, Aug. 2004.
[43] J. Shi and J. Malik, "Normalized cuts and image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 22, no. 8, pp. 888–905, Aug. 2000.
[44] D. Huang, C.-D. Wang, J.-S. Wu, J.-H. Lai, and C.-K. Kwoh, "Ultra-scalable spectral clustering and ensemble clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 6, pp. 1212–1226, Jun. 2020.
[45] R. Wang, F. Nie, Z. Wang, F. He, and X. Li, "Scalable graph-based clustering with nonnegative relaxation for large hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7352–7364, Oct. 2019.
[46] X. Peng, Z. Yu, Z. Yi, and H. Tang, "Constructing the L2-graph for robust subspace learning and subspace clustering," *IEEE Trans. Cybern.*, vol. 47, no. 4, pp. 1053–1066, Apr. 2017.
[47] M. Shin and M. Brbic, "Selective sampling-based scalable sparse subspace clustering," in *Proc. 33rd Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2019, pp. 12416–12425.



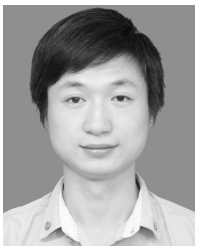
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